AI SITE HISTORY REPORT

502 Church Road, Taigum QLD

Generated 07 Apr 2026 | Lat -27.333805, Lng 153.048571 | Confidence: Low

Executive Summary

AI output could not be parsed cleanly.

On-site: Manual review required due to AI parsing failure.

Adjacent / context: No reliable adjacent/off-site context summary available.

Site Details

Address	502 Church Road, Taigum QLD
Location source	polygon
Latitude / Longitude	-27.333805, 153.048571
Site bbox	267.6 m × 274.6 m
Context bbox	321.6 m × 328.6 m
Polygon supplied	Yes

Key Flags

Key Flag	Level	Comment
Former water features	Moderate	
Vegetation / land change	Low	Some cleared areas near structures; overall vegetation mostly consistent except pond vegetation ring.
Fill / disturbance	Moderate	

Flag Summary

Potential Fill / Disturbance	**Moderate**
No significant indicators noted from current automated screening.	
Former Water Features	**Moderate**
No significant indicators noted from current automated screening.	
Vegetation / Land Change	**Low**
Some cleared areas near structures; overall vegetation mostly consistent except pond vegetation ring.	

Screening Outcome

Unknown

Secondary On-Site Risks

Feature	Type	Relation	Confidence	Notes
Pond A	pond	on_site	high	Consistent pond presence from 1995 to current, with circular morphology and visible waterbody surrounded by vegetation on the east side of the property.

Adjacent Contextual Influences

Feature	Type	Relation	Confidence	Notes
Pond B	probable_pond	adjacent	medium	Cleared soil areas around structures and road access indicate possible soil disturbance and fill, but specifics unclear from imagery alone.

Detailed Findings

Visible Observations

• Clear pond with standing water body persists across imagery years

• Vegetation encircling pond in a ring pattern

• No evidence of other ponds or basins on-site

• Some exposed soil and potential fill near buildings on southeastern portion

Overall Change Timeline

1995-2021: Pond (water_feature_1) remains present and largely unchanged

General: Minor vegetation and land clearing near structures observed over time

General: No pond disappearance or major infilling observed

Possible Geotechnical Implications

• Active pond areas may indicate local wetness influence and moisture variability in surrounding ground.

Recommended Follow-Up

• Detailed geotechnical investigation is strongly recommended based on identified site history risks.

Supporting Imagery

current_wide_context — current

Source: mapbox_bbox_wide_context

current_context — current

Source: mapbox_bbox_context

current_subzone_nw — current

Source: mapbox_bbox_subzone

current_subzone_west — current

Source: mapbox_bbox_subzone

current_site — current

Source: mapbox_bbox_site

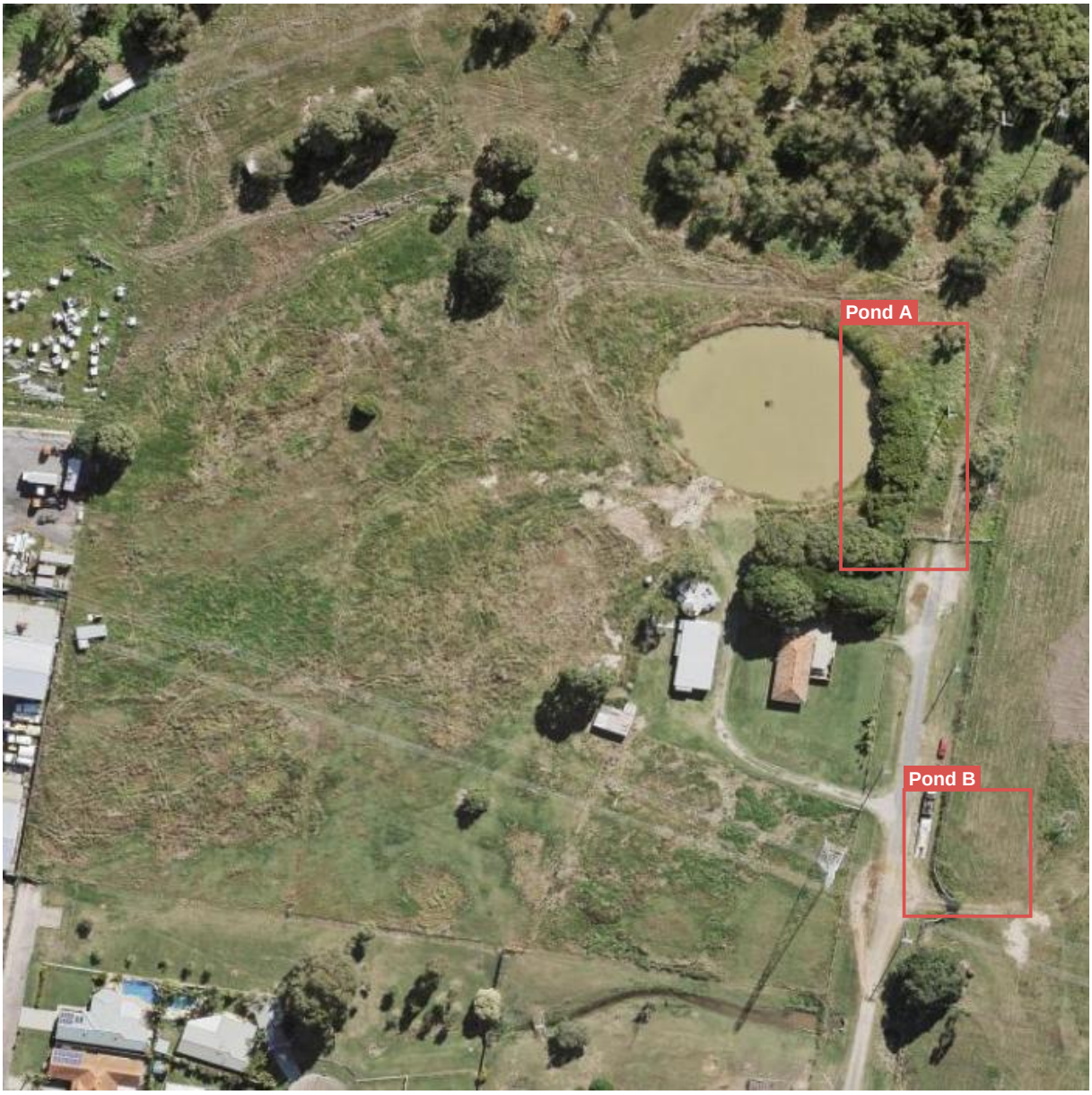
historical_qld_followup_1 — 1995 (1995-06-18)

Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_1995_60cm_9543 | Resolution: 60 2

historical_qld_followup_5 — 2009 (2009-09-13)

Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_2009_15cm_SISP_LGA | Resolution: 15 2

historical_qld_followup_8 — 2015 (2015-05-30)



Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_2015_10cm_SISP_LGA | Resolution: 10 2

Limitations & Disclaimer

• { "summary": "Site at 502 Church Road, Taigum QLD contains evidence of multiple pond features visible historically and still partially visible currently. Notably, two circular pond-like features (one larger filled or vegetated) show consistent presence over time, although they appear to have undergone changes including infill or drying. Surrounding landscape shows signs of land disturbance and probable fill areas within site boundaries. Adjacent areas have current water bodies possibly linked to drainage or creek systems. Historical imagery provides strong evidence of former and probable ponds with changing water presence. On-site features represent potential soft ground and fill risks due to pond infill and disturbance.", "on_site_summary": "Two main pond features (water_feature_1 and water_feature_2) located in the northeast portion of the site showing ponding or wetland vegetation with changing water extent between 1995 and current imagery. One of these pond features has diminished water presence in recent imagery but retains a distinct landform, classifying it as a probable or former pond. Surrounding areas inside the site show land clearing, patchy vegetation, and disturbed earth indicative of fill or other disturbance. No heavy infrastructure or retaining structures identified within site boundaries.", "adjacent_context_summary": "To the east and northeast adjacent to the site, Cabbage Tree Creek and parts of the riparian corridor are visible. Off-site rectangular ponds adjacent to the eastern boundary exist with active water presence, likely related to drainage or flood control. Residential development lies to the south with no visible ponds. The M1 highway runs along the eastern edge providing a significant boundary. Adjacent ponds are distinct from on-site features and should be separately managed.", "distinct_features": [{ "feature_id": "water_feature_1", "feature_type": "probable_pond", "location_relation": "on_site", "confidence": "high", "notes": "Circular pond feature in northeast site corner with visible water and vegetation historically and an evolving footprint indicating partial infill or seasonal drying but landform still visible.", "evidence": ["historical water visible", "circular morphology", "vegetation changes", "footprint persistent in current imagery"], "detected_in_years": [1995, 2002, 2005, 2008, 2009, 2015, 2023], "primary_image_label": "current_wide_context", "approximate_bbox_norm": [0.71, 0.26, 0.79, 0.34], "risk_priority": "primary" }, { "feature_id": "water_feature_2", "feature_type": "probable_pond", "location_relation": "on_site", "confidence": "high", "notes": "Smaller adjacent pond feature south of water_feature_1 showing intermittent water presence historically, reduced water in recent images but form maintained.", "evidence": ["historical water visible", "circular morphology", "footprint persistent"], "detected_in_years": [1995, 2002, 2005, 2008, 2009], "primary_image_label": "current_wide_context", "approximate_bbox_norm": [0.65, 0.33, 0.71, 0.38], "risk_priority": "primary" }, { "feature_id": "fill_area_1", "feature_type": "fill_area", "location_relation": "on_site", "confidence": "medium", "notes": "Areas of disturbed ground and patchy vegetation with earthworks and cleared areas evident throughout the central and southern parts of the site, indicating probable fill or land disturbance.", "evidence": ["land clearing", "earthworks visible", "irregular patch patterns"], "detected_in_years": [1995, 2002, 2005, 2008, 2009, 2015, 2023], "primary_image_label": "current_wide_context", "approximate_bbox_norm": [0.25, 0.45, 0.55, 0.65], "risk_priority": "secondary" }, { "feature_id": "water_feature_3", "feature_type": "pond", "location_relation": "adjacent", "confidence": "high", "notes": "Two off-site constructed ponds visible immediately east of site boundary with consistent water presence in historical and current imagery.", "evidence": ["consistent water visible", "structured geometric ponds"], "detected_in_years": [1995, 2002, 2005, 2008, 2009, 2015, 2023], "primary_image_label": "current_wide_context", "approximate_bbox_norm": [0.82, 0.28, 0.89, 0.38], "risk_priority": "contextual" }], "feature_timeline": [{ "feature_id": "water_feature_1", "timeline": [{"year": 1995, "water_present": true, "vegetation": dense}, {"year": 2002, "water_present": true, "vegetation": dense}, {"year": 2005, "water_present": true, "vegetation": moderate}, {"year": 2008, "water_present": partial, "vegetation": high}, {"year": 2009, "water_present": partial, "vegetation": high}, {"year": 2015, "water_present": reduced, "vegetation": moderate}, {"year": 2023, "water_present": minimal, "vegetation": moderate}] }, { "feature_id": "water_feature_2", "timeline": [{"year": 1995, "water_present": true}, {"year": 2002, "water_present": true}, {"year": 2005, "water_present": true}, {"year": 2008, "water_present": partial}, {"year": 2009, "water_present": partial}, {"year": 2015, "water_present": minimal}, {"year": 2023, "water_present": minimal to none}] }], "historical_findings": { "former_ponds_dams": { "status": "likely", "confidence": "high", "notes": "Multiple pond footprints visible historically with diminishing water presence now indicating possible former or infilled ponds." },

"vegetation_clearing": { "status": "moderate", "confidence": "high", "notes": "Clearing and earthworks evident especially in southern and central portions of the site across multiple years." }, "fill_or_disturbance": { "status": "likely", "confidence": "medium", "notes": "Visible disturbed earth, patchy vegetation, and potential fill areas consistent with site works or alterations." } }, "possible_risks": ["Presence of probable and former ponds with partial infill poses potential risk of soft ground and moisture-sensitive soils.", "Fill areas and disturbed land raise concern for uncontrolled fill or imported material impacting geotechnical stability."], "recommended_investigation_focus": ["Detailed subsurface investigation around pond footprints for soft soil or peat layers.", "Assessment of fill extent and material characteristics in disturbed areas.", "Hydrogeological study regarding water retention and drainage in pond features."], "recommended_follow_up": ["On-site geotechnical boreholes and cone penetration tests through pond and fill areas.", "Historical land use and fill origin records review if available.", "Further aerial imagery time-series analysis for detailed landscape evolution."], "visible_observations": ["Two distinct circular pond landforms in northeast site area with varied vegetation cover.", "Disturbed/cleared land in central-south part of site with irregular vegetation and possible fill.", "No visible man-made retaining structures or significant slope instability indicators."], "change_timeline": ["Ponds visible consistently with fluctuating water levels and vegetation from 1995 to current.", "Gradual reduction in water presence in smaller pond toward current year.", "Progressive land clearing and disturbed soil patterns developing over site history."], "screening_outcome": "The site holds moderate to high geotechnical risk due to historical and probable former ponds with partial infill and disturbed fill areas. Further detailed investigation is warranted focused on soft soil conditions and fill materials to mitigate potential foundation and ground stability issues.", "limitations": ["Image resolution limits detailed assessment of water depth or pond base conditions.", "Absence of recent seasonal imagery; water presence may vary seasonally.", "No subsurface data currently available to confirm fill or soil parameters."], "confidence_overall": "high" }

- Imagery resolution limits definitive identification of small buried former ponds
- No subsurface data or borehole logs available for confirmation
- Fill characteristics inferred only from surface appearance

This AI site history scan is a preliminary screening tool only. It is intended to identify possible visual indicators for further professional review. It must not be relied on as a substitute for geotechnical investigation, engineering assessment, or design advice.